



Review

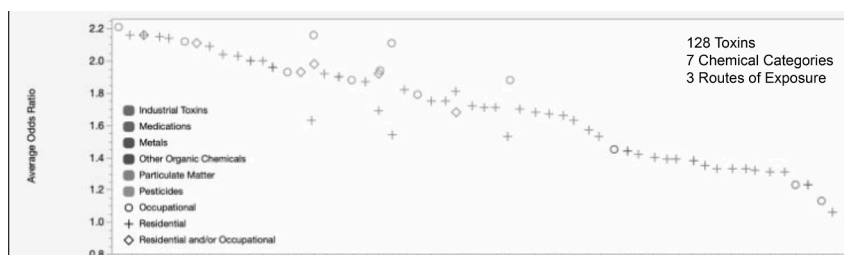
Literature review and meta-analysis of environmental toxins associated with increased risk of Parkinson's disease

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HIGHLIGHTS

- Environmental toxins linked to Parkinson's Disease (PD) are examined.
- Occupational exposures to toxins are significantly associated with PD.
- Industrial toxins and food contaminants are important risk factors.
- Persistent toxins accumulated in fatty food (whale meat) are PD risk drivers.
- Data gaps are identified for geographic PD prevalence rates globally.

GRAPHICAL ABSTRACT



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ABSTRACT

Parkinson's disease (PD) is a neurodegenerative disorder and leading cause of death worldwide, whose pathogenesis has been linked to toxic environmental exposures. We used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (i) to compile, and group by exposure setting (non-specified general; residential; occupational), environmental factors reported to modulate the risk of developing PD and (ii) to map and geospatially analyze global regions of both research activity and paucity. Among the broader environmental settings, occupational exposures had the highest average odds ratio value at 3.82, followed by general (non-specified or mixed) exposures at 3.07, and residential exposures at 2.36. Occupational exposure to industrial toxins was the highest ranked subset of exposures with an odds ratio of 10.74. Among the studies meeting the inclusion criteria, 75 % were conducted in Europe or the Western United States. The number of individuals partaking per study ranged from a high of 55,585 (Taiwan) to a low of 233 (Faroe Islands), with a mean of $n = 14,462$. The top three environmental factors associated with high odds ratios for increased risk of developing PD were (i) exposure to dyes (25.33), (ii) methylene chloride (16.5) and specifically in adult men (iii) consumption of fatty whale meat (10.57), which is known to harbor a broad spectrum of so called persistent, bioaccumulative, toxic (PBT) pollutants. Geospatially, the highest odds ratio values were identified in European countries, whereas notable data gaps were revealed for South America, Australia, Africa, and the majority of Asia

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with the exception of Taiwan. Whereas occupational exposures to industrial chemicals, such as harmful dyes and methylene chloride, ranked highest in risk values, available data suggest notable opportunities for reducing PD cases globally by limiting harmful environmental exposures to a spectrum of toxic chemicals, particularly via the food intake route. Thus, current efforts in improving environmental quality globally by limiting toxic emission may deliver the added benefit of helping to reign in PD. Agents of concern in this respect include pesticides (e.g., paraquat, demeton, monocrotophos), particulate matter associated with air pollution, and a spectrum of organic and inorganic neurotoxins including heavy metals.

1. Introduction

Parkinson's disease (PD), an ailment linked to both genetic predisposition and harmful environmental factors, cut short 329,000 lives worldwide in 2019 (World Health Organization, 2022), was the 14th leading cause of death in the US (Halli-Tierney et al., 2020), and currently is the second most common neurodegenerative disorder after Alzheimer's as well as the most common age-related movement disorder in the United States and in European nations (Mhyre et al., 2012). Onset of PD is characterized by the progressive destruction of dopaminergic neurons in the substantia nigra pars compacta and, similar to other neurodegenerative diseases, cases often present with Lewy bodies, or clusters consisting primarily of the α -synuclein protein (Bereczki, 2010). The most common clinical manifestations of the disease, i.e., resting tremors, rigidity, bradykinesia, and postural instability, are consistent with findings that implicate the role of these dopaminergic neurons in fine motor control and coordination (Triarhou, 2002), although a number of symptoms unrelated to these functions may warrant further study in changes associated with other neurons (Lang and Obeso, 2004). It is estimated that about 40 % of cases are linked to heritable genetic mutations (Wirdefeldt et al., 2011). Overall, mutations in the GBA gene are the most common risk factors for PD (Brockmann and Berg, 2014). Other mutations in both familial and sporadic cases of PD include the LRRK2 protein (Kluss et al., 2019; Kumari and Tan, 2009; Rivero-Ríos et al., 2020). Classic presentations of PD are generally autosomal dominant and linked to mutations in genes such as GBA, SNCA and LRRK2 (Li et al., 2014; Kessler et al., 2018), producing characteristic late-onset PD with the aggregation of α -synuclein Lewy bodies. However, forms of slower, earlier-onset PD without the presence of Lewy bodies have been identified and linked to dysregulation of multiple autosomal recessive genes, notably PARK, PINK1 and DJ1 (Trempe and Fon, 2013).

However, while genetics play a significant role in the onset of PD, this alone cannot explain current findings in the field of epidemiology. Environmental factors appear to have considerable influence in the pathogenesis of PD, possibly serving as a trigger to genes involved in the disease. Research has attempted to identify toxins and environmental settings which have statistically significant associations with disease incidence. Currently, there are numerous compounds that show some correlation to the pathogenesis of the disease. For example, the effect of pesticide exposure, such as that of Paraquat, has been extensively studied in clusters in the Central California region (Tanner et al., 2011; Berry et al., 2010). A range of compounds, including metals, industrial solvents (i.e., chlorinated hydrocarbons), particulate matter in air pollution, and other neurotoxic chemicals have been implicated, notably aflatoxin B1 and ochratoxin A, both common mycotoxins that appear to affect oxidative stress pathways in specific parts of the brain and nervous system (Sava et al., 2006; Gugliandolo et al., 2020). Furthermore, lifestyle factors such as the use of antidepressants or anxiolytic medications, and even being rendered unconscious may have some link to developing PD (Dick et al., 2007). However, despite the existence of a still expanding list of toxic environmental factors and lifestyle choices associated with PD, clear causal links are often difficult to establish, as is the case for many disorders triggered by the environment.

Environmental influence may vary due to regional idiosyncrasies in the presence and concentration of toxins. To gain a better understanding

of the effect of population-wide toxic exposures on disease incidence rates and differences in pathogenesis, it is imperative to consider case-control and cohort studies from various regions across the world. Furthermore, geospatial epidemiological studies can provide a more robust analysis of toxins associated with the disease by accounting for location-dependent differences in affected populations.

The present literature review and meta-analysis examined environmental factors known or suspected to be associated with an increased risk of developing PD. It specifically examined general and distinct environmental exposure settings as well as composited and distinct environmental factors linked to the disease. Furthermore, we mapped the geographic location of cohorts of individuals that have been investigated thus far to identify data gaps in the literature.

2. Methods

2.1. Literature analysis

A comprehensive literature search was performed by obtaining articles from databases including Scopus. The Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) Statement was used to guide the literature search and determine which articles would be included in the review (Supplemental Fig. 5). Of one thousand, two-hundred and twenty-two articles identified, a total of twelve articles were selected for inclusion in this meta-analysis. Fifty-one articles were assessed for eligibility, and thirty-nine were excluded for failing to meet the inclusion criteria (Supplemental Fig. 5).

A systematic literature review was conducted on Scopus for articles available on or before January 2022. Parkinson's was searched for using {Parkinson's} OR {Parkinsonism} OR {Parkinsons}. Toxins were selected for using the {Tox*} OR {poison} OR {contaminant}. To filter for studies with subjects who were known to be exposed to the toxin(s), {External} OR {Exogenous} OR {Exposure} OR {incidence} was used. Finally, to include geospatial studies, the keywords {Geo*} OR {mapping} OR {region*} were used.

The inclusion criteria for each study was as follows: (1) Peer-reviewed studies published in English from January 2000 to December 2021 (Supplemental Fig. 4); (2) consisted of case-control, cohort or retrospective studies in which subjects/controls included at least 50 individuals each, as a larger sample size would better represent the populations being studied; (3) clearly identified chemical exposures with statistically significant correlations to Parkinson's disease.

While the keywords cast a wide net in order to collect the broadest yield, the exclusion criteria carefully eliminated studies that did not meet the inclusion criteria. A majority of studies were excluded from this manuscript for the following reasons: Studies were performed before January 2000; Data drawing correlations between acute or chronic chemical exposures seen in animal models and effects of environmental exposures in human populations; experimental methods were not case-controlled, cohort-based, or hypothesis-driven.

2.2. Geospatial analysis

The ArcGIS Pro 3.0 geographic information system software was used to construct geospatial maps showing the regions referenced in studies in this meta-analysis. The purpose of this was to observe whether

certain regions or countries had higher OR values in comparison to other regions.

2.3. Meta-analysis

For each exposure category, an average odds ratio was calculated by averaging the published odds ratios categorized by each study. For example, the “Averaged OR” regarding residential exposure was calculated by averaging all published odds ratio values categorized by each paper as a “residential exposure”. To determine the likelihood of a Type I error (false positive), a relative population weight (RPW) was calculated by dividing the population size of each study by the median sample size in the literature reviewed ($n = 2756$). A relative population weight above 1.0 suggests a lower likelihood of a Type 1 error caused by a relatively small sample size. Therefore, this value is intended to strengthen trust towards OR values from studies with larger sample sizes, under the assumption that a larger sample size reduces the potential that a data set was observed by chance.

3. Results

3.1. Literature review results

Application of stringent inclusion criteria resulted in a subset of research studies selected from an initial count of 1222 works considered featuring data for 3 continents, 10 countries, 3 exposure settings and 128 contaminants of both organic and inorganic composition (See SI for additional information).

3.2. Occupational exposures

For occupational exposures, we observed the highest average OR value at 3.82. Individually, factors in the occupational category ranked extremely high among all factors listed. The top four factors in this

category – dyes, methylene chloride, naphtha and derivatives, and demeton – displayed OR values of 25.33, 16.5, 9.53 and 4.74, respectively. Only one other factor – residential men’s whale meat consumption in adulthood at 10.57 – exceeded that of naphtha and derivatives. On the other hand, occupational exposure to fungicides (1.45 OR), quartz dust (1.23 OR), and detergents (1.13 OR) produced a low average OR value of 1.27, despite the fact that both quartz dust and detergents are within the highest ranked subcategory of industrial toxins (Figs. 1, 4).

3.3. Occupational and residential exposures

Factors within the residential and/or occupational exposure category, when averaged, had the second highest OR value at 3.07. methamidophos, monocrotophos, oxydemeton-methyl and demeton are the top four factors in this division, and are all organophosphate pesticides, with OR values at 4.72, 5.53, 5.86 and 5.93, respectively. The four factors with the lowest OR values appear were methidathion, exposure to 1–7 organophosphate pesticides (OPs), phosmet, and chlorpyrifos, at 2.11, 1.98, 1.93 and 1.92, respectively (Fig. 3). All factors within this category are pesticides, and all factors except for one come from a singular study.

3.4. Residential exposures

Factors listed as residential exposures had the lowest OR value of all three exposure types at 2.36. The top three factors were men’s whale meat consumption in adulthood, men’s blubber consumption in adulthood, and women’s whale meat consumption in adulthood. The factors with the lowest three OR values were river or well water exposure at 1.23, nitrogen dioxide at 1.06, and manganese exposure at 1.034 (Fig. 2).

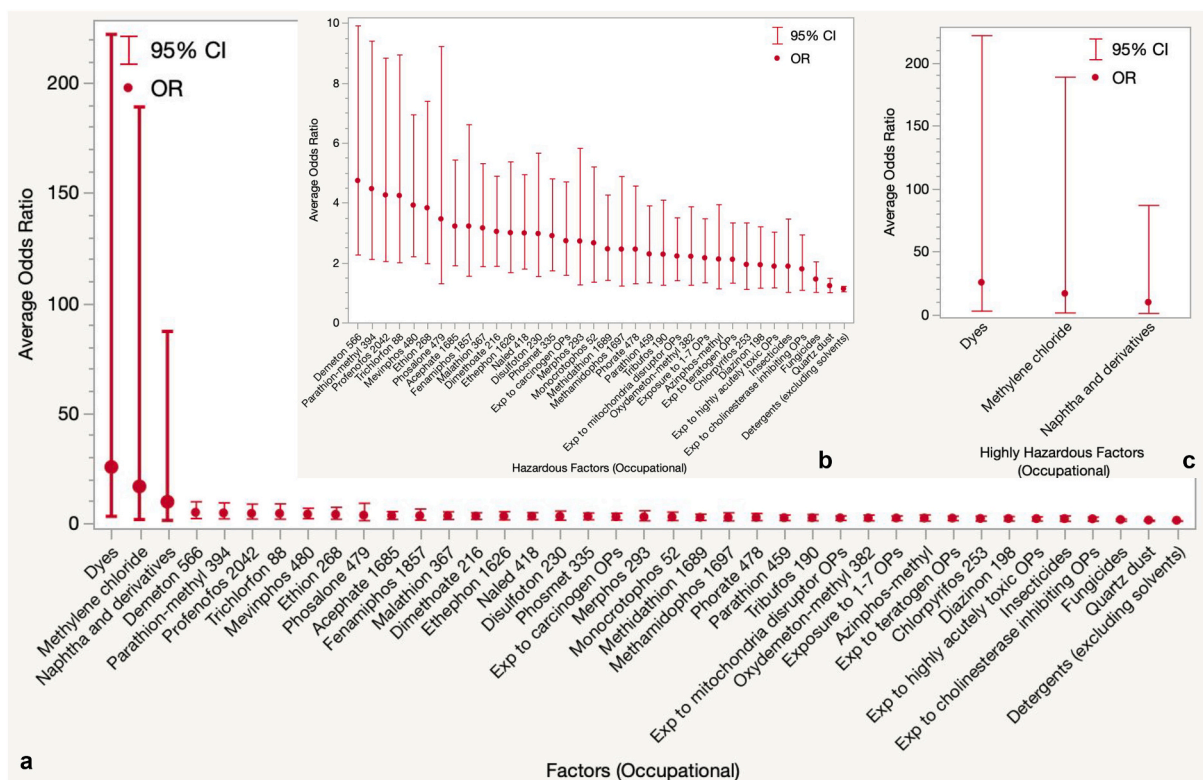
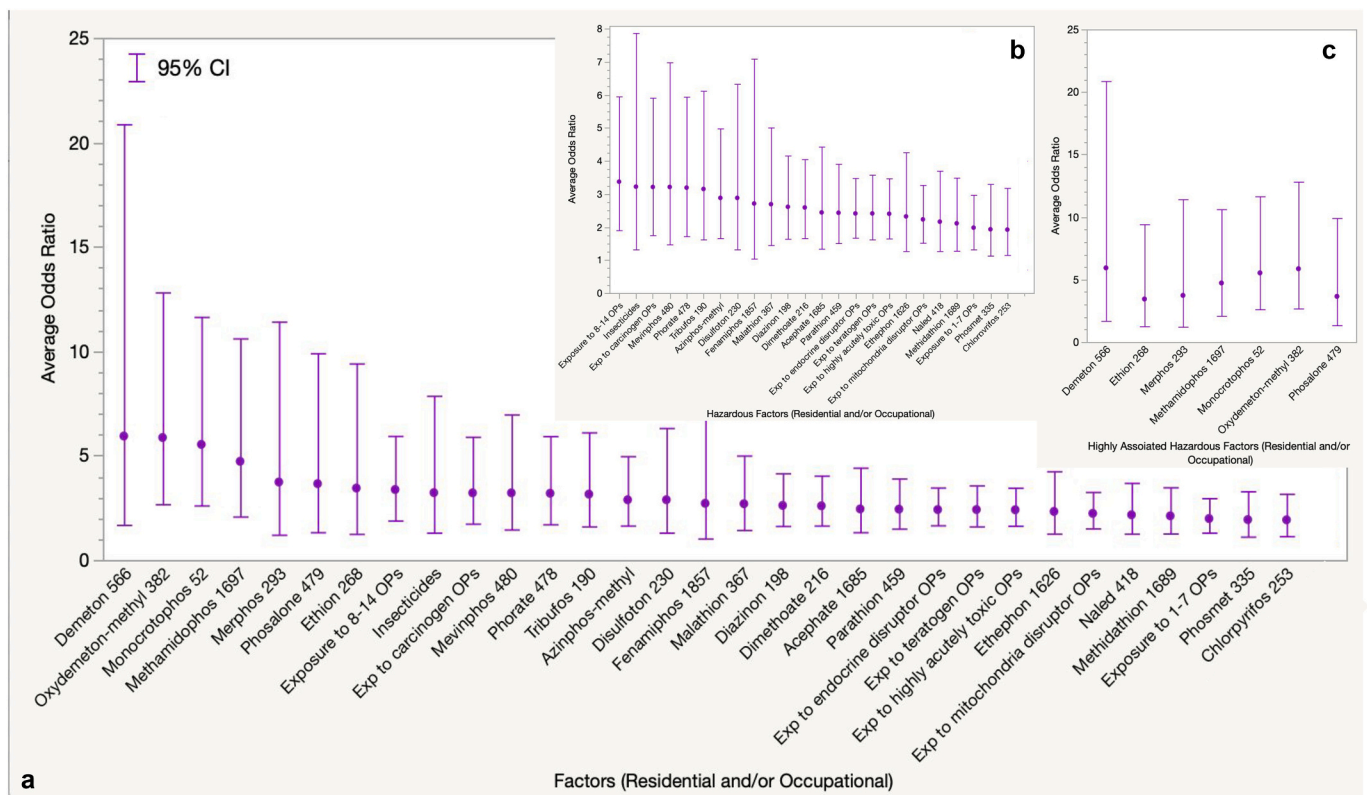
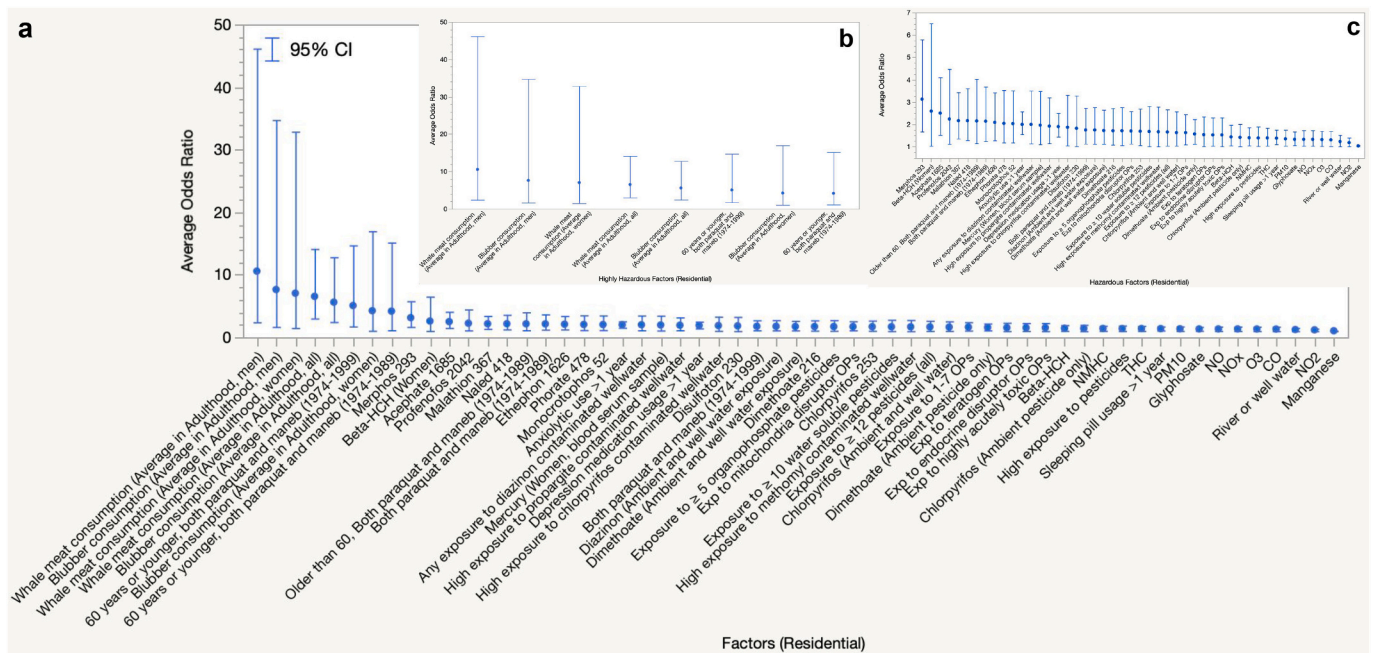


Fig. 1. Factors associated with Parkinson's Disease (exposure in occupational setting) and respective average odds ratio (OR) values, reported from the selected studies/factors ranked by descending odds ratio value.



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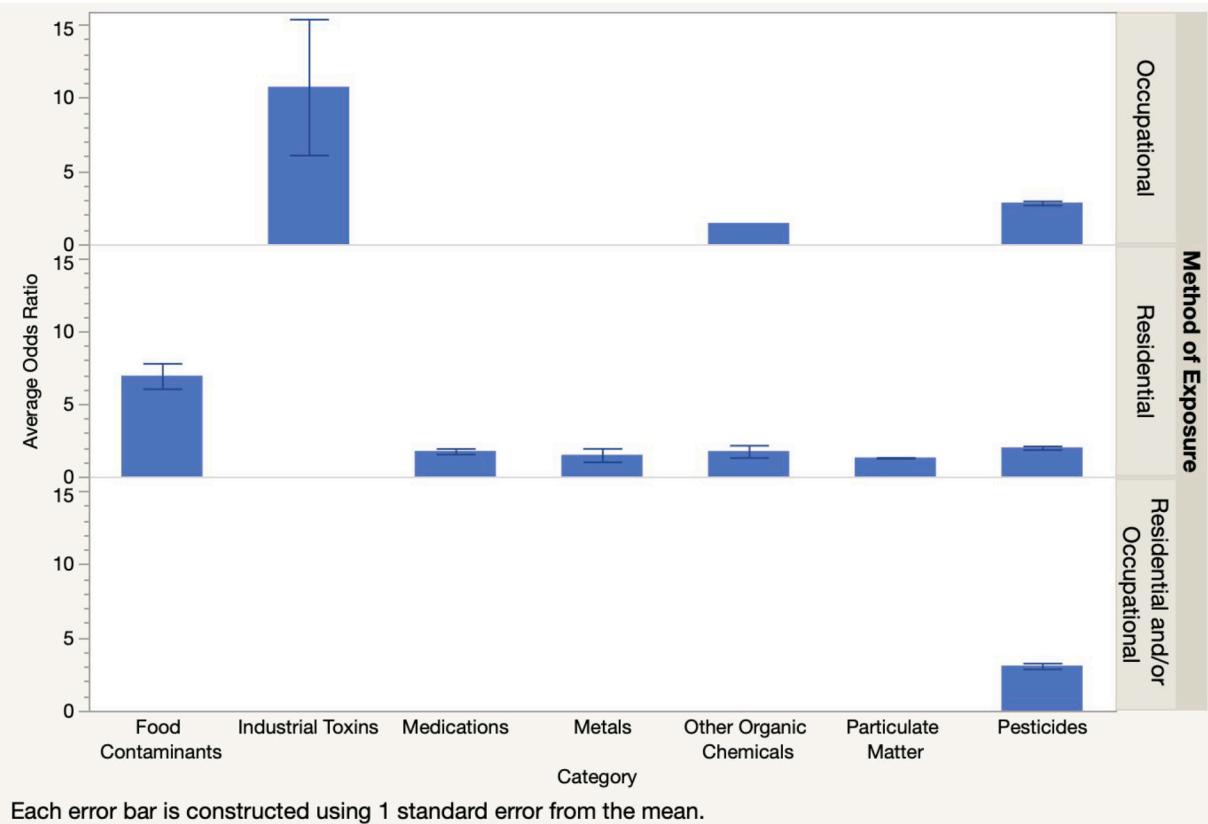


Fig. 4. Mean odds ratio values of factors grouped by type and exposure. Mean of average OR values reported from the selected studies/factors versus list of factors associated with increased chance of developing Parkinson's Disease in all settings, grouped by type of toxin as well as method of exposure (residential, occupational, or both).

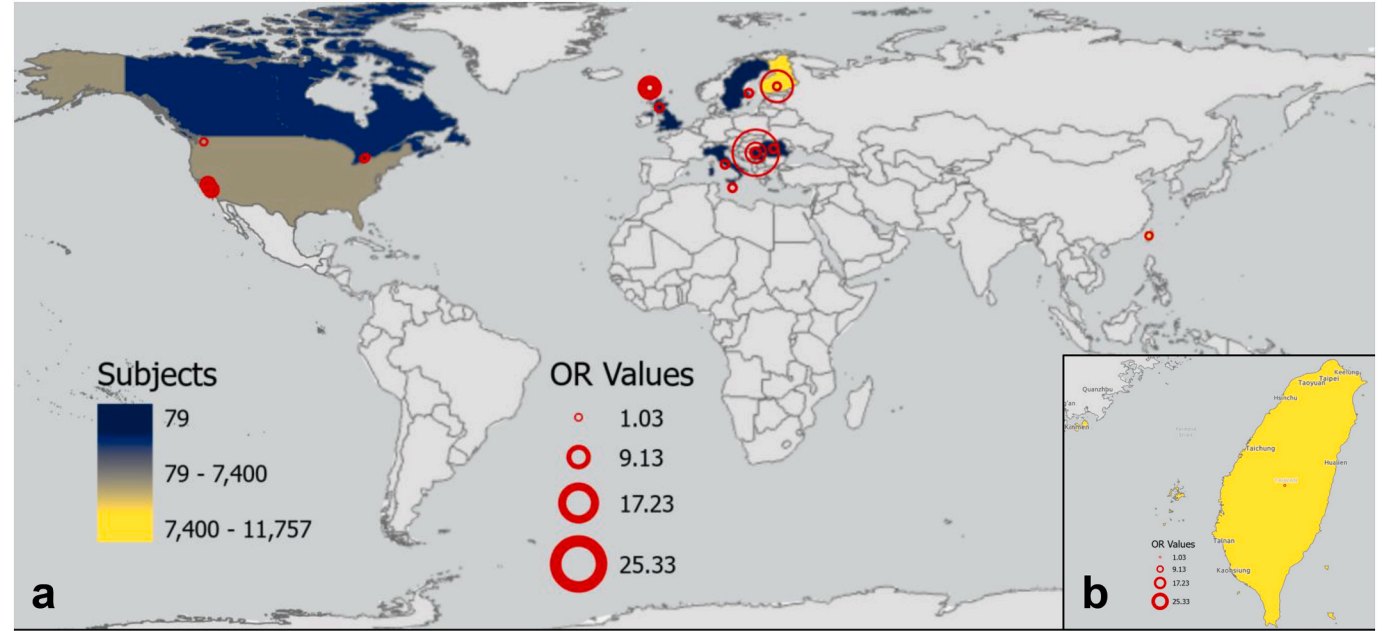


Fig. 5. Geographic mapping of odds ratio values and sample size. Odds ratio values of toxin exposures associated with Parkinson's Disease reported by population-based exposure studies are plotted by sample size geographically using ArcGIS. The size of the red circles corresponds to odds ratio values. The blue to yellow color spectrum represents the number of subjects in each study. A larger circle indicates a higher OR value.

3.6. Relative population weight comparisons

Out of the studies selected for this analysis, eight studies, selected

based on their relevance to the meta-analysis (e.g. high OR values, highly ranked subcategories), along with their respective OR values and the relative population weight (RPW) have been provided for

comparison (Supplementary Table 1). Now, when comparing the top four factors- dyes (RPW 0.12), methylene chloride (RPW 12.7), men's whale meat consumption in adulthood (RPW 0.08), and naphtha and derivatives (RPW 0.12)- every factor except for methylene chloride (Nielsen et al., 2021) comes from studies with relative population weights below 1.0, a result of these studies all having small sample sizes.

3.7. Geographic distribution

Some 75 % of the studies included in this review were conducted in the United States or European countries (Fig. 5, Supplemental Fig. 1). The data in the three studies excluded from this category were collected in Canada and Taiwan (Fig. 5, Supplemental Fig. 1). All studies considered were case-control or cohort studies, while a single experiment utilized a retrospective cohort approach (Fig. 5, Supplemental Fig. 1). The average cohort size was 14,462 individuals. However, this value does not reflect the wide range of sample sizes within these studies, which were likely formed to reflect population size. The smallest sample group was 233 individuals from the Faroe Islands (Supplemental Fig. 2), while the largest included 55,585 individuals from Taiwan (Supplemental Fig. 2). Of the top 4 factors with the highest OR values – exposure to naphtha and derivatives at 9.53, men's whale meat consumption in adulthood at 10.57, methylene chloride at 16.5 and dyes at 25.33 – all aside from whale meat consumption are associated with occupational exposures, and were identified in Finland and Serbia (Supplemental Fig. S2).

4. Discussion

Overall, occupational exposures – particularly the exposure to food contaminants and pesticides – were some of the most highly ranked factors with correlations suggestive of links to PD (Supplemental Fig. 3). When sorted according to environmental exposure setting, occupational exposures had the highest average OR value at 3.82. However, it is unclear where these exposure events may have occurred, as the subjects' occupations and the number of years spent in an occupation were not stated in many of the studies. One paper does provide a table with similar information for both subjects and controls, but it is unclear whether a greater length of time spent in a single occupation correlates with increased risk of PD due to a dosing effect (Vlajinac et al., 2010). Furthermore, study-specific information is missing for various factors; for example, it is unclear what derivatives of naphtha were studied, and the same can be said about “dyes” as a factor.

Another limitation of this work is that exposures could not always be linked to specific chemicals. A number of toxins included in this analysis – such as persistent bioaccumulative toxins found in dietary intake of marine food, or the derivatives of naphtha mentioned above – are considered to be complex chemical matrices. Detailed data regarding their chemical composition could shed more light on their role in the pathogenesis of PD as well as accumulation patterns in soil, air, or contaminated food products. The latter would be especially useful for toxins that are typically involved in residential exposure events.

The high value associated with occupational exposures may be due to three of the four toxins with the highest OR values – dyes, naphtha and derivatives, and methylene chloride – being categorized as occupational exposure to industrial toxins (Fig. 4). This likely explains a much larger average OR value of 10.74 associated with occupational exposure to toxins in industrial settings. Residential and/or occupational exposures ranks second at an average OR value of 3.07, with food contaminants representing the highest ranked subcategory at 6.94. Lastly, residential exposures had the lowest average OR value at 2.36. However, data for the residential and/or occupational exposures category in particular is incomplete, given that all but one exposure within this category come from a single study. The category itself is rather subjective, and determining where the majority of these exposures occurred is difficult. With regards to the study that composes almost the

entirety of this category, this study states that there is a clear separation between subjects exposed in residential addresses, occupational addresses, or at both residential and occupational addresses (Wang et al., 2014). For the purpose of this paper, however, this category has been modified as “residential and/or occupational” for two reasons: because subjects whose workplaces are within 500 m of agricultural fields (where pesticides have been applied) are considered ambiently exposed, and because data regarding the subjects' occupations is not available (Wang et al., 2014). With that being said, the study does provide additional comparisons and odds ratios based on the pesticides' mechanisms of action, which may provide more insight into specific pathways affected in the pathogenesis of Parkinson's and whether certain toxins may result in a more rapid onset or progression of the disease.

The highest OR values in the residential category were found for consumption of whale meat. Although this diet certainly is rare, the underlying exposures most certainly are not. Whale fat accumulates a broad spectrum of environmental pollutants scavenged by the large animal from the oceans over its long lifetime. Whale meat is rich in fat in which persistent organic pollutants (POPs) become trapped and concentrated over time. Hence, the findings for whale meat suggest that prolonged human exposure to complex mixtures of POPs may well be an important driver of risk for developing PD. Specific environmental toxins having been detected at elevated concentrations in whale meat include but are not limited to congeners of dioxins, polycyclic chlorinated biphenyls (PCBs), polybrominated diphenyl ethers as well as persistent pesticides such as DDT and its breakdown products (DDE) (Petersen et al., 2008; Fängström et al., 2005).

When organized into qualitative categories, slight discrepancies from the overall trends emerged. For example, exposure to other organic chemicals in residential settings, with an average OR value of 2.01 ranks higher than occupational exposure at 1.45. These results may be due to the effects of ambient exposure compounded over decades, even though it is generally assumed that acute exposures in workplace settings carry higher risks. However, residential and/or occupational exposure to pesticides had an OR value of 3.07, followed by occupational exposure to pesticides at 2.84, and residential exposure at 2. Further research is needed, as certain categories (such as metals and nearly all pesticide exposures, for example) are based on data from a small number of studies. Further research in other regions and/or countries is needed to establish regional differences in toxin exposure and to draw comparisons between other toxins that may not be present in the locations currently being studied. Incremental exposure data to measure absorption of these chemicals is also unavailable. However, one study included in this analysis separates exposures by time windows based on subjects' residential addresses at the estimated time of exposure as well as age. The study's findings suggest that exposure to toxins (in this case, pesticides) in an earlier time period or at a younger age are associated with a higher risk of developing PD (Costello et al., 2009). Similar data in future studies is necessary to better differentiate between acute and ambient, chronic exposures in various exposure settings, which may affect the acceleration of pathogenesis in affected populations. Since the physiological symptoms of PD are generally observed in older age, exposure data grouped by both time and age could be useful in determining whether the early development of disease may occur earlier than previously assumed. More detailed comparisons are difficult to make due to the lack of data in the following categories: occupational exposure to food contaminants, residential and/or occupational exposure to food contaminants, residential exposure to industrial toxins, residential and/or occupational exposure to industrial toxins, occupational exposure to medications, residential and/or occupational exposure to medications, occupational exposure to metals, residential and/or occupational exposure to metals, residential and/or occupational exposure to other organic chemicals, occupational exposure to particulate matter, and residential and/or occupational exposure to particulate matter.

Other discrepancies can be seen when examining the Type 1 error analysis. For example, the Petersen et al., 2008 study, whose OR values

make up the entirety of the food contaminants subcategory, has a sample size of just 233 individuals, and the lowest RPW of 0.08. Both dyes and naphtha and derivatives also come from a study with a comparatively smaller sample size and a RPW of 0.12. The same can be said of the Gatto et al., 2009 (RPW 0.26), Wang et al., 2014 (RPW 0.4), and Costello et al., 2009 (RPW 0.26) studies, which account for the majority of pesticide exposures. However, the methylene chloride value (OR 16.5), which is derived from the Nielsen et al., 2021 study conducted in Finland, is associated with an RPW of 12.7, implying that there is a low probability of a Type I error. Given that the compounds in this study make up a significant portion of the industrial toxins subcategory, there is evidence to further suggest that occupational exposure to chemicals in this subcategory have a significant correlation with PD. On the other hand, the particulate matter subcategory, which was the lowest ranked subcategory within this analysis, contains values from the Chen et al., 2017 and Lee et al., 2016 studies, the latter of which has the largest sample size ($n = 55,585$) of all the studies included in this analysis. With RPW values of 1.84 and 20.17, respectively, it appears that toxins categorized as particulate matter may warrant further study, especially in areas showing increased urbanization (Chen et al., 2017).

However, it is worth noting that the data provided cannot account for significant regional differences between studies. For example, pesticides have been extensively studied in California, given that its economy is highly dependent on agriculture. Any relationship between pesticides and chronic illnesses would have major implications and/or consequences for the millions who work in affected industries. Unfortunately, the heterogeneity across the studies in this analysis poses an obstacle in making comparisons between exposures. For example, the studies included in this analysis conducted in European countries focus on industrial toxins and a limited number of organic chemicals, such as fungicides. Where pesticides were included, it was not specified whether these were organophosphates or similar in chemical composition to the pesticides examined in the studies from the United States. All data regarding air particulate matter comes from Asia, more specifically, Taiwan. Lastly, the toxins studied by Nielsen et al. in the Faroe Islands appear to be specific to this populations' diet, which is not shared by the other populations or regions studied in this analysis. Given the chemical compositions of the exposures studied in each region vary widely, and their ubiquity may be specific to location, it is unclear whether these toxins warrant further study globally.

Even within the same region, studies conducted across different time periods may yield different results, depending on the current dispersion or concentration of toxins being studied. Interactions between these compounds and their environments have not been thoroughly studied, in part because it is nearly impossible to account for the thousands of chemical reactions that may occur depending on what medium the compound resides in. Due to these reactions, concentrations may change over time, which would affect how widespread exposure events are in each decade. There is also the possibility that these reactions may produce other compounds that may lead to detrimental effects in exposed populations. Of course, the interactions mentioned do not nullify current findings. That being said, future epidemiological research that studies these changes would provide a more thorough understanding on the direct and indirect impacts of the exposures being studied.

In addition, smaller populations by region may not be able to obtain large population sizes given local constraints. One study mentioned multiple times in this analysis, conducted in the Faroe Islands, has the smallest sample size at just 233 individuals, and focuses on contaminated food products, namely whale meat and blubber, which are not commonly consumed worldwide (Petersen et al., 2008). As a result, the compounds in this study may be deemed less significant compared to studies conducted in other countries with a much larger population and/or cohort size. However, it must be acknowledged that whale meat and blubber are a staple food source for individuals living in the Faroe Islands. Further studies regarding the surrogate mixture of the persistent bioaccumulative toxins found in whale meat and other marine food can

still be informative to this particular population as well as other regions with high marine food consumption.

With regards to individual studies, it should be noted that two papers included in this analysis utilize the same study population. With that being said, the exposure criteria and the toxins examined in each study varies somewhat. More specifically, while the study conducted by Wang et al. focuses on organophosphate pesticides and considers proximity to agricultural land in determining exposure (Wang et al., 2014), the Gatto et al. study focuses on well water to determine exposure, and as a result, also examines pesticides that are not considered organophosphates but have a high potential for runoff and leaching, therefore having an increased chance of contaminating well water (Gatto et al., 2009).

A further limitation of these studies was the ambiguity about the exposures both with respect to a clear distinction of occupational versus residential and temporary, with respect to long-term and short-term exposures. Each exposure has been categorized as occupational, residential, or residential and/or occupational based on details provided by each study. Given discrepancies between categorizations in each study, however, and that occupational and residential environments are not isolated from one another, it is difficult to separate occupational from residential exposures or from mixed exposures, especially those resulting from both environments, as well as clearly defining an exposure as acute or chronic.

Future research should fill data gaps extant outside of the United States and the European nations (Fig. 5). Future work also may aim to identify at-risk populations residing in presently under-studied regions. Sample sizes need to be increased to improve the quality of information obtained and the strength of conclusions drawn. However, differences in the type of study may lead to inconsistent results. For example, since case control studies select individuals that have already been identified as having PD, the OR values associated with exposure to compounds are based on prior exposures. Although PD can take several decades to develop, and therefore, OR values based on prior exposures are of high importance, this may not account for the effect of more recent exposures in exacerbating disease. More data is needed to better understand associations between acute or short-term exposures and their effects on PD. Incremental data that directly measures toxin concentrations in air and soils near residential and occupational spheres over several years would provide data on transmission levels in populations. With regards to mixed exposures (residential and/or occupational exposures), the geographical criteria used to define a population as residentially exposed, occupationally exposed, or exposed in both settings must be better defined, especially as it pertains to subjects residing near, but not working in, occupational zones with known toxin use. Similarly, subjects' length of residence in the region being studied is needed to account for dosage effects. Although the length of residency should be a limiting factor in selecting subjects for a study, this limit should be adjusted given the half-lives, excretion rates, and varying toxicities of the compounds of interest, all of which affect the duration of the exposure. Following this logic, data regarding subjects' occupations and length of occupation could serve as a means of better identifying the toxins and the concentration they may have been exposed to. Although biopsies from individuals with confirmed diagnoses in these regions could establish stronger links between certain toxins and the progression of PD, performing invasive biopsies on individuals who are currently alive may be impractical and potentially unethical. However, analyzing post-mortem biopsies may be an alternative way to establish causal relationships between environmental factors and their association to PD. With that being said, increased data regarding transmission could be utilized to improve population-based public health interventions to screen for and aid susceptible communities.

The epidemiological evidence in this paper reveals various gaps in our knowledge as well as ambiguities in certain factors. Geospatial studies linking environmental exposure to PD incidence are concentrated to a relatively small number of countries, notably the United States and Europe. Similarly, on the national and state level, a high

percentage of studies in California are often limited to three counties in the Central Valley, and many of these studies focused almost exclusively on pesticides or exposure to contaminated well water, whereas studies in Europe are generally concentrated on industrial exposures but are considerate of residential factors as well. The lack of case studies performed in other countries leads to incomplete reporting, especially since there may be undiscovered factors unique to these regions. One study included in this paper formed sample sizes based on a retroactive cohort that assumes exposure to certain factors due to location and/or diagnosis (Caballero et al., 2018). Also, the majority of studies in this analysis pooled participants from hospitals or other clinical centers. These case-control studies were based upon physicians' diagnosis of PD in subjects, or extended use of medications used to treat the symptoms of PD, such as Levodopa or Carbidopa (Finkelstein and Jerrett, 2007). Therefore, existing and future geospatial studies could be hindered due to any discrepancies in records or lack of representation within sample sizes. It is also worth noting that this review may be incomplete as articles written in languages other than English were not considered for this analysis.

Although case control or cohort studies in regions with high incidence of PD may provide links between environmental factors and risk of developing the disease, different studies appear to produce conflicting results. For example, studies have identified contaminated well waters, such as those containing Propargite or Methomyl, to be associated with an elevated risk of developing PD (Gatto et al., 2009); however, a study surveying Medicare beneficiaries across the United States that suggested an inverse correlation when exposed to well water (Silver et al., 2020). Moreover, the aggregation of quantifiable factors into qualitative categories, such as particulate matter, metals or heavy metals, pesticides, manganese, etc., makes it difficult to quantify levels of exposure. These ambiguities also make it difficult to determine whether certain exposure events occurred in residential or occupational settings. Furthermore, case control or cohort studies that lack quantitative risk statistics cannot be included as they do not have the specific data with which to make comparisons between other studies. While this does not invalidate observed associations, this means exact exposure routes and duration of exposures may never be known.

In summary, occupational exposures, particularly those involving industrial toxins or pesticides, were most significantly associated with PD. This was followed by residential and/or occupational exposures (in which pesticides were the only subcategory), then residential exposures (with food toxins being the highest ranked subcategory). Within individually ranked toxins, occupational exposure to dyes had the highest OR value at 25.33, and residential exposure to manganese had the lowest OR value at 1.034. Based on this evidence, further research should conduct cohort studies in regions currently underrepresented in literature to better identify global trends in toxin exposure events. Given that exposure to many of these toxins is essentially unavoidable, it is imperative that both these hazardous chemicals and effects of certain routes of exposure are further studied. Screening and other preventative measures could thereby be implemented for communities determined to be at higher risk of developing PD as a result of occupation, dietary factors, and region of residence.

CRedit authorship contribution statement

Anumitha Aravindan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **Melanie Engstrom Newell:** Writing – review & editing, Conceptualization. **Rolf U. Halden:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.scitotenv.2024.172838>.

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